BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 119 · Random Numbers and Combined Constructs

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does SELECT specify in an SQL query?

Q2. What does a higher CPU clock speed mean?
A. More installed cores
B. More bytes per pixel
C. More cycles per second
D. More storage per file

Q3. “Larger cache increases the number of installed CPU cores.” Correct this. Explain why.

Q4. CPU A: 4 GHz, two cores, 2 MB cache. CPU B: 3 GHz, eight cores, 8 MB cache. Why is neither fastest for every task?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Random integer — A pseudorandom function can select a whole number from a stated inclusive range when specified.

Sequence — Statements execute one after another unless control flow changes.

Selection — A condition decides whether a branch runs.

Iteration — A loop repeats a process while or until an appropriate stopping condition is met.

New learning

Random integer: A pseudorandom function can select a whole number from a stated inclusive range when specified.

Sequence: Statements execute one after another unless control flow changes.

Selection: A condition decides whether a branch runs.

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Worked example

Trace a dice game: roll=6; if roll>=5 print('Win'), else print('Try again').

1. Identify the given roll 6.

2. Evaluate 6 >= 5 as true.

3. Choose the winning branch.

4. The output is Win.

We Do · Practise with guidance

Guided 1

A simulated roll is 3 and the program wins only on 6. Predict the output.

Worked method: The output is Try again.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the bonus-message output when the dice roll is six.

```
roll = random(1,6)
if roll == 6 then
    print(...)
else
    print("Keep trying")
endif
```

write

Question 1

Explain the purpose of random number generation in a dice simulation. [2 marks]

Question 2

State two programming constructs in a game with a loop and IF statement. [2 marks]

Question 3

Why should a game test both the minimum and maximum permitted dice rolls? [2 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A condition can determine which branch of a program runs.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The fields (columns) to return.

Q2. C — More cycles per second

Q3. A larger cache holds more often-used data and instructions. This can reduce slower memory access.

Q4. A has more cycles per second, which may help one task. B has more cores for parallel work and more cache to reduce slow memory access. Speed also depends on the task and CPU design.

I Do / We Do pages · Guided practice

Q1. The output is Try again.

You Do pages · Code First

```
roll = random(1,6)
if roll == 6 then
    print("Bonus!")
else
    print("Keep trying")
endif
```

Expected result: Bonus! if roll is 6; otherwise Keep trying

You Do pages · Independent questions

Q1. It models different possible outcomes rather than a fixed predetermined roll.

Q2. Iteration and selection.

Q3. They are boundaries of the expected range and may expose off-by-one errors.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: